

CS5004, Spring 2026

Reading and Storing Data from a File

Objective

The objective of this lab is to practice reading data from a text file and storing it in objects. You will read a file containing student information (first name, last name, student ID, and email) line by line. Each student's data will be stored in a *Student* object and added to an *ArrayList*.

Problem 1: Prepare the Text File

Create a **students.txt**. Each line in this file contains a student's details in the following format:

FirstName LastName StudentID Email

Add the following lined to your text file:

John Doe 12345 john.doe@example.com

Jane Smith 67890 jane.smith@example.com

Mark Johnson 11223 mark.johnson@example.com

Emily Davis 44556 emily.davis@example.com

Problem 2: Implement the Java Program

Write a Java program that performs the following tasks:

Reads the **students.txt** file line by line.

Extracts the *first name, last name, student ID, and email* using appropriate methods.

Stores each student's data in a *Student* object.

Adds each *Student* object to an *ArrayList*.

Loops over the *ArrayList* and prints each student's details, including their corresponding line number.

Displays a menu to the user asking the user to choose:

1. Add a student:
 - a. which will prompt the user for the information about the new student.
 - b. Add an object to represent the new student
 - c. Update the students.txt file
 - d. Print the new list of students
2. Remove a student by specifying the ID number:
 - a. Which will delete the object corresponding to the chosen student.
 - b. And delete the information about that student from the students.txt file.
3. Search for a student by ID number:
 - a. Which will display the information about the student with the required ID number if the student exists in the database
 - b. Display an error message indicating the student does not exist

Problem 3: Submit the Following Files

- **StudentReader.java** – The program that reads the file and processes the data.
- **Student.java** – The class representing a student.
- **students.txt** – The text file containing student information.